

Improving Edge AI Efficiency With High-Accuracy In-Memory Computing

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Introduction

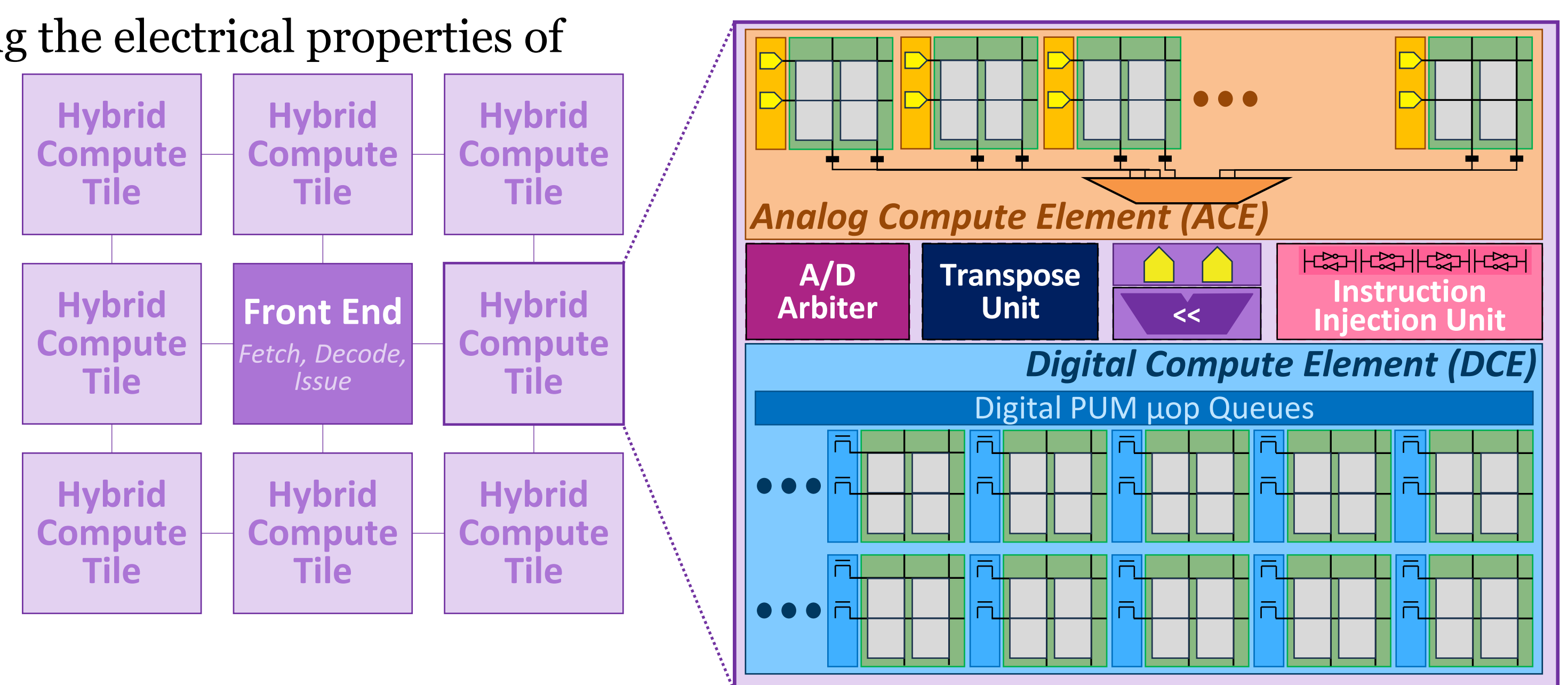
- Immersive applications hurt by **data handling inefficiencies**
 - Existing computer HW **wastes significant energy/perf.**
 - Tight resource constraints @ edge
- Today, many operations are infeasible in edge constraints
 - Training: too much energy
 - Complex analytics: need for always-on network
- In-memory computing can make **adv. local compute** possible

Other Recent Projects

- Memory Processing Unit:** a front-end abstraction for PUM to enable a universal software stack [Truong+ HPCA 2026]
- BRIE-RAM:** biodegradable PUM hardware for sustainable compute [Das+ HotEthics 2026]
- ANVIL:** in-SSD bulk search operations for name-value data structures [Wong+ ISCA 2025]

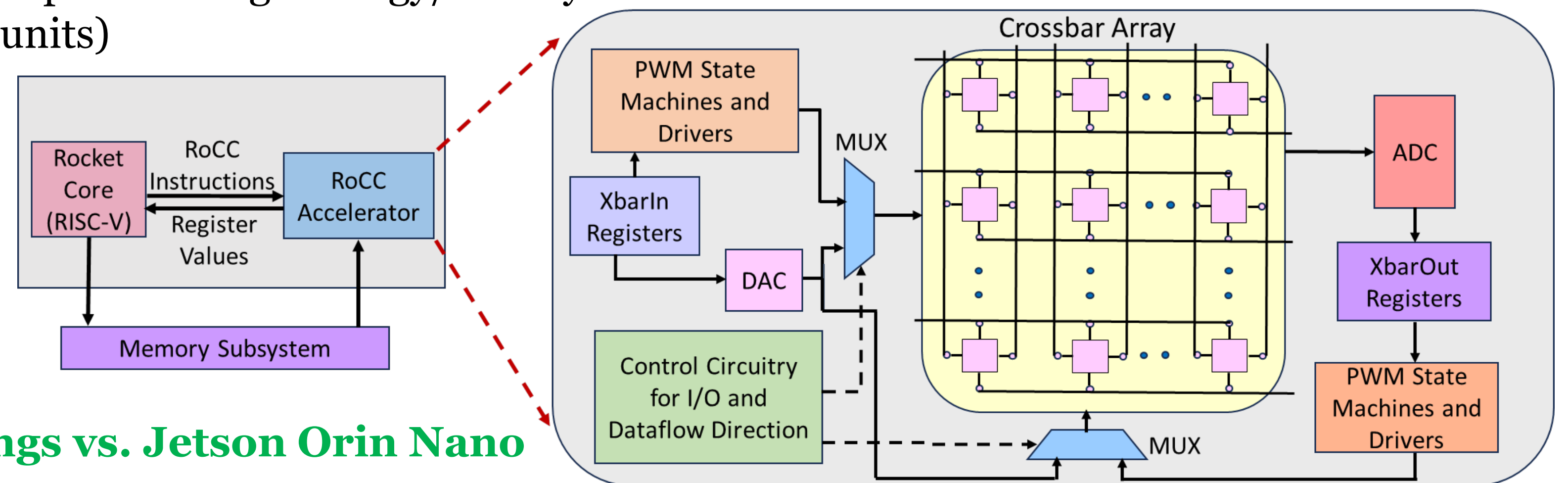
DARTH-PUM: Performing All Compute in Memory w/ Hybrid Analog+Digital Operations [Wong+ ASPLOS 2026]

- Processing-using-memory (PUM)** can enable computation by using the electrical properties of memory devices for useful inter-device interactions
 - Analog PUM primarily used for *matrix-vector multiplication* (MVM)
 - Still requires constant offload to the CPU for non-MVM ops
- DARTH-PUM: Digital-Analog Resistive Tiles for Hybrid PUM
 - Same ReRAM tiles/arrays, two modes: best of both worlds**
 - Digital PUM performs bulk bitwise Boolean logic
 - Hardware optimizations handle synchronization and domain conversion between mixed-signal logic boundaries
- Evaluation averaged over ResNet-20, AES, and LLM encoding: **11.8x performance improvement, 7.5x energy savings** vs. NVIDIA RTX 4090 GPU



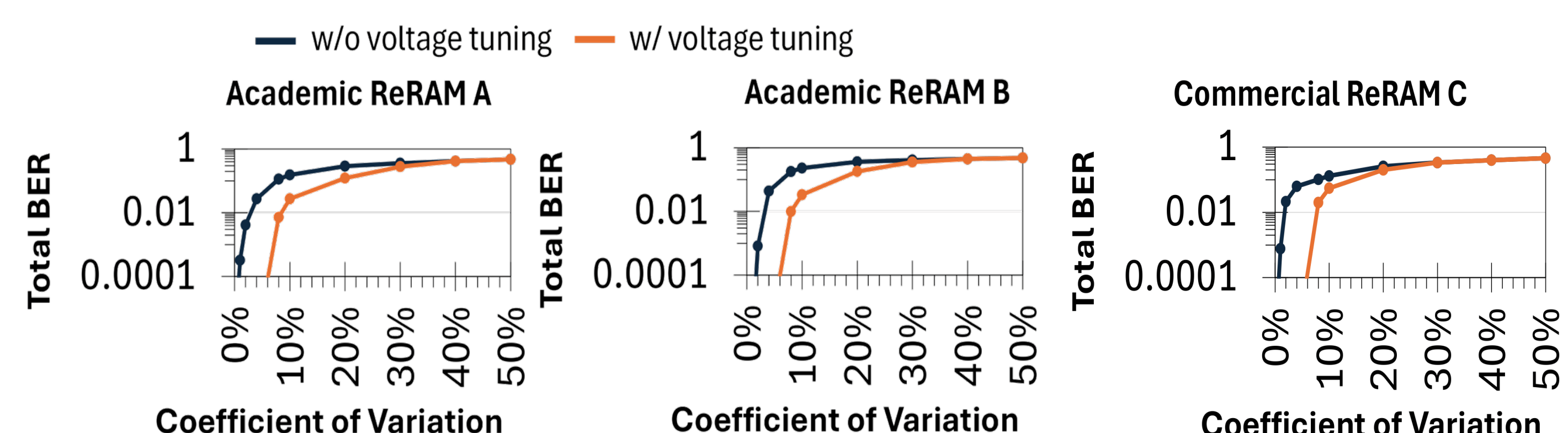
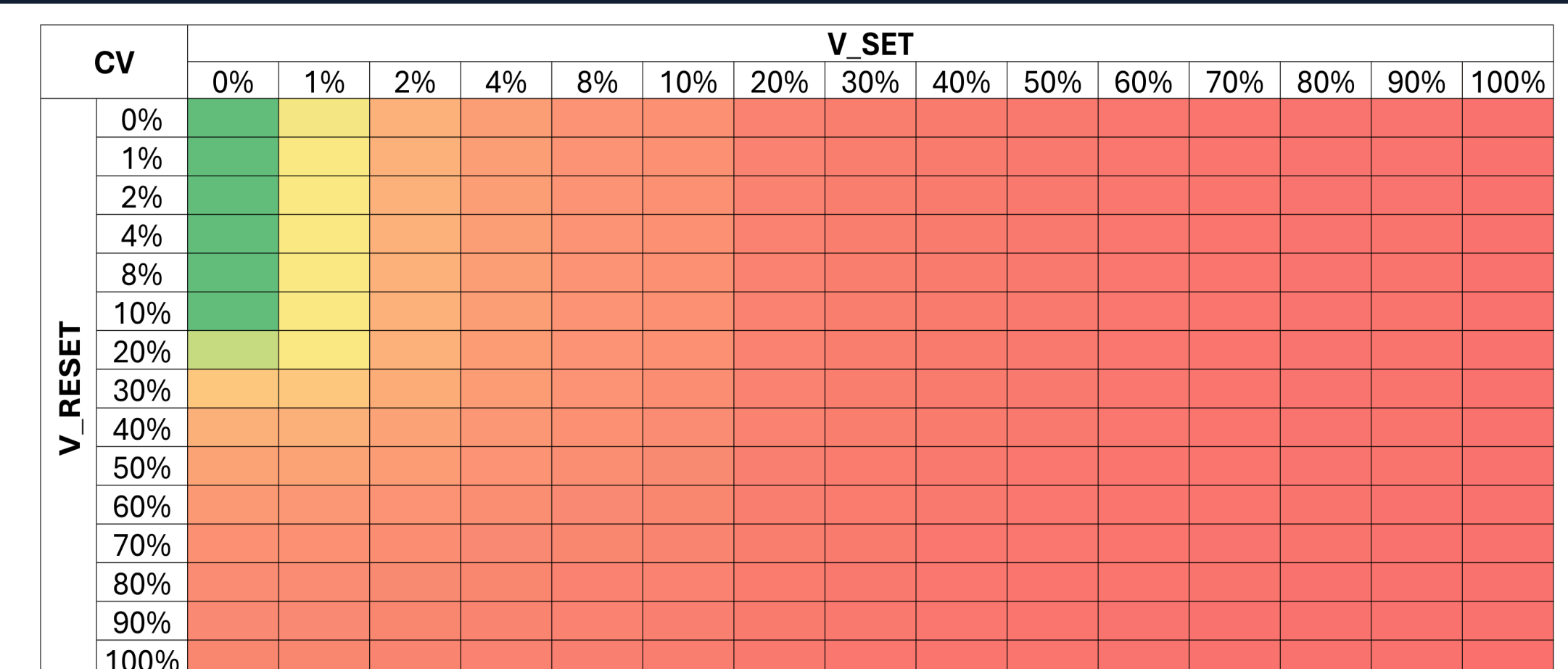
Enabling In-Edge Continual Learning w/ New Memory Technologies [Tasnim+ AICS 2025]

- Continual learning (CL):** a solution for ML models to get updated incrementally without losing old pre-trained information, even without having to keep all the original data around
- State-of-the-art CL algorithms optimized for CPU/GPU, but experience high energy/latency (main contributor: data movement b/w memory & compute units)
- New memory: **electrochemical RAM (ECRAM)**
 - Linear and symmetric programming characteristics
 - Ultra-low device-to-device / cycle-to-cycle variability
- CLASP: Continual Learning Acceleration System Platform
 - ECRAM (fabbed in house) made into IMC accelerator
 - RoCC interface to RISC-V core using custom instructions
- Eval w/ 5 CL algs: **up to 92% accuracy, 51x energy savings vs. Jetson Orin Nano**



Architecture Lessons for Improving PUM Reliability [under submission]

- We have developed MASTODON, a cross-stack simulator to study the functionality of emerging memory devices and their operation in a chip-scale memory architecture
 - Enables early-stage design space exploration using real-world applications to drive experiments across the entire stack
 - ReRAM: 600+x speedup vs. SPICE, <2% error of transients
- We are utilizing MASTODON to **make digital PUM more reliable**
 - No real data today on bit error rates (BERs) of Boolean PUM
 - Using new stochasticity-encoded device models, we're exploring circuit-level & architectural-level techniques to make PUM reliable enough for general compute (BER < 10⁻¹²)
- Voltage tuning, optimized triple-modulo redundancy (TMR) show strong potential in reducing BER
- Currently working on broader techniques for app-specific correction



Acknowledgments

- Work supported in part by NSF FuSe (CCF-2329096), Sandia National Labs LDRD, UIUC DREMES HYBRID Center, UIUC ASAP Center (an NSF IUCRC), Intel

- Thanks to our collaborators: Ben Feinberg (Sandia); Qing Cao, Haoran Liu, Shaloo Rakheja, Siyuan Qian, Golam Samdani, Shomik Chatterjee (UIUC); Divake Kumar, Amit Trivedi (UIC)
- Papers available at <https://arcana.cs.illinois.edu/>

